

Supplementary Tables

Table 1: Performance comparison for 4mC.

Dataset	Model	ACC	SN	SP	MCC	AUC	F1
4mC_ <i>C.equisetifolia</i>	BiLSTM	77.87	76.50	79.24	55.76	87.25	77.56
	CRNN	78.14	73.22	83.06	56.56	87.07	77.01
	TextCNN	75.68	72.68	78.68	51.46	86.18	74.93
	Transformer	74.32	62.30	86.34	50.10	82.11	70.81
	Ultimate_Fusion_NetWork	84.97	83.06	86.88	70.00	90.42	84.68
4mC_ <i>F.vesca</i>	BiLSTM	84.35	87.05	81.65	68.80	91.74	84.76
	CRNN	84.65	87.52	81.78	69.42	92.28	85.08
	TextCNN	71.80	75.36	68.24	43.72	79.07	72.77
	Transformer	84.79	85.33	84.25	69.59	92.50	84.87
	Ultimate_Fusion_NetWork	86.11	86.46	85.76	72.22	93.45	86.16
4mC_ <i>S.cerevisiae</i>	BiLSTM	58.39	62.18	54.60	16.83	63.25	59.91
	CRNN	60.97	61.78	60.16	21.94	66.32	61.28
	TextCNN	59.86	49.75	69.97	20.13	64.44	55.34
	Transformer	69.62	59.76	79.48	40.02	77.02	66.29
	Ultimate_Fusion_NetWork	71.99	63.20	80.78	44.68	78.46	69.29
4mC_ <i>Tolypocladium</i>	BiLSTM	59.31	62.83	55.79	18.67	63.06	60.70
	CRNN	71.16	72.41	69.91	42.33	77.81	71.52
	TextCNN	58.38	59.01	57.75	16.77	61.85	58.64
	Transformer	73.66	69.84	77.48	47.46	81.63	72.61
	Ultimate_Fusion_NetWork	75.58	73.55	77.61	51.21	83.53	75.08

Table 2: Performance comparison for 6mA.

Dataset	Model	ACC	SN	SP	MCC	AUC	F1
6mA_ <i>A.thaliana</i>	BiLSTM	84.67	81.10	88.24	69.52	92.19	84.11
	CRNN	85.27	83.52	87.02	70.58	92.90	85.00
	TextCNN	71.69	67.88	75.50	43.51	78.82	70.57
	Transformer	85.77	82.84	88.70	71.67	93.07	85.34
	Ultimate_Fusion_NetWork	86.52	84.58	88.46	73.10	93.86	86.25
6mA_ <i>C.elegans</i>	BiLSTM	85.20	87.29	83.11	70.46	92.47	85.50
	CRNN	88.91	89.77	88.05	77.83	95.41	89.00
	TextCNN	72.86	73.97	71.75	45.74	80.68	73.16
	Transformer	89.97	89.87	90.07	79.95	96.08	89.96
	Ultimate_Fusion_NetWork	91.63	93.47	89.79	83.32	97.08	91.78
6mA_ <i>C.equisetifolia</i>	BiLSTM	60.70	65.38	56.02	21.49	64.57	62.46
	CRNN	67.38	65.18	69.58	34.78	73.13	66.64
	TextCNN	59.96	63.50	56.42	19.96	64.82	61.33
	Transformer	72.78	67.62	77.94	45.81	80.06	71.30
	Ultimate_Fusion_NetWork	74.88	72.63	77.13	49.80	82.45	74.30
6mA_ <i>D.melanogaster</i>	BiLSTM	89.97	88.44	91.50	79.98	95.65	89.82
	CRNN	91.23	89.88	92.58	82.50	96.33	91.11
	TextCNN	76.32	76.09	76.55	52.64	83.97	76.26
	Transformer	90.71	89.85	91.57	81.42	96.42	90.63
	Ultimate_Fusion_NetWork	92.68	91.87	93.49	85.37	97.32	92.62
6mA_ <i>F.vesca</i>	BiLSTM	85.78	84.53	87.03	71.59	92.55	85.60
	CRNN	91.52	91.04	92.00	83.05	96.53	91.48

Table 2 – Continued from previous page

Dataset	Model	ACC	SN	SP	MCC	AUC	F1
6mA_H.sapiens	TextCNN	79.11	82.40	75.82	58.35	87.99	79.78
	Transformer	91.97	94.00	89.94	84.02	97.13	92.13
	Ultimate_Fusion_NetWork	92.75	93.29	92.21	85.50	97.74	92.79
	BiLSTM	88.68	85.61	91.75	77.51	95.72	88.32
	CRNN	90.04	88.63	91.45	80.11	96.53	89.90
6mA_R.chinensis	TextCNN	79.94	78.84	81.04	59.89	88.57	79.72
	Transformer	90.19	88.08	92.30	80.45	96.54	89.98
	Ultimate_Fusion_NetWork	90.76	89.04	92.48	81.57	96.93	90.60
	BiLSTM	67.06	62.54	71.58	34.25	72.90	65.50
	CRNN	68.56	62.21	74.91	37.43	76.90	66.43
6mA_S.cerevisiae	TextCNN	73.75	72.24	75.26	47.51	79.60	73.34
	Transformer	81.94	82.94	80.94	63.89	89.52	82.12
	Ultimate_Fusion_NetWork	81.77	78.60	84.94	63.67	90.53	81.17
	BiLSTM	68.46	62.44	74.48	37.20	74.89	66.44
	CRNN	78.26	72.69	83.83	56.88	86.60	76.98
6mA_T.thermophile	TextCNN	67.64	59.48	75.80	35.77	74.66	64.77
	Transformer	80.30	76.44	84.16	60.77	88.48	79.51
	Ultimate_Fusion_NetWork	82.44	76.12	88.76	65.39	90.47	81.25
	BiLSTM	88.31	95.25	81.37	77.36	94.51	89.07
	CRNN	88.53	95.19	81.87	77.75	94.67	89.24
6mA_Tolypocladium	TextCNN	67.68	68.54	66.82	35.36	74.33	67.95
	Transformer	87.78	94.97	80.59	76.35	94.12	88.60
	Ultimate_Fusion_NetWork	88.49	95.01	81.97	77.65	94.75	89.20
	BiLSTM	62.34	64.89	59.79	24.72	67.41	63.28
	CRNN	64.92	71.76	58.08	30.12	71.32	67.17
6mA_Xoc BLS256	TextCNN	63.26	68.03	58.49	26.65	69.88	64.93
	Transformer	77.03	71.05	83.01	54.45	84.79	75.57
	Ultimate_Fusion_NetWork	77.47	80.82	74.12	55.07	86.05	78.20
	BiLSTM	85.37	85.56	85.18	70.75	92.79	85.40
	CRNN	87.44	86.52	88.36	74.89	94.50	87.32
	TextCNN	70.74	65.69	75.79	41.70	78.36	69.19
	Transformer	87.41	87.10	87.72	74.82	94.55	87.37
	Ultimate_Fusion_NetWork	88.90	88.24	89.56	77.80	95.70	88.83

Table 3: Ablation comparison for 4mC.

Dataset	Model	ACC	SN	SP	MCC	AUC	F1
4mC_C.equisetifolia	No_BiLSTM	74.59	65.57	83.61	50.00	83.77	72.07
	No_Freq	78.14	74.86	81.42	56.41	85.64	77.40
	No_Time	77.05	72.13	81.97	54.36	84.23	75.86
	No_Transformer	78.42	73.77	83.07	57.08	86.83	77.36
	Ultimate_Fusion_NetWork	84.97	83.06	86.88	70.00	90.42	84.68
4mC_F.vesca	No_BiLSTM	83.41	86.57	80.25	66.96	91.35	83.92
	No_Freq	84.78	87.72	81.84	69.68	92.33	85.22
	No_Time	84.62	86.63	82.61	69.30	92.33	84.93
	No_Transformer	83.88	85.83	81.93	67.82	91.54	84.19
	Ultimate_Fusion_NetWork	86.11	86.46	85.76	72.22	93.45	86.16
4mC_S.cerevisiae	No_BiLSTM	69.77	67.14	72.40	39.59	75.22	68.95
	No_Freq	67.34	70.58	64.10	34.75	73.36	68.36
	No_Time	68.71	74.12	63.30	37.63	75.76	70.31
	No_Transformer	68.30	54.30	82.30	38.13	75.21	63.14
	Ultimate_Fusion_NetWork	71.99	63.20	80.78	44.68	78.46	69.29
4mC_Tolypocladium	No_BiLSTM	72.82	77.88	67.76	45.87	80.40	74.13
	No_Freq	74.12	77.89	70.35	48.38	82.35	75.06
	No_Time	74.03	74.38	73.68	48.06	81.75	74.12
	No_Transformer	73.74	76.81	70.67	47.58	81.74	74.53
	Ultimate_Fusion_NetWork	75.58	73.55	77.61	51.21	83.53	75.08

Table 4: Ablation comparison for 6mA.

Dataset	Model	ACC	SN	SP	MCC	AUC	F1
6mA_A.thaliana	No_BiLSTM	84.49	83.73	85.25	68.98	92.15	84.37
	No_Freq	85.15	82.75	87.55	70.38	92.76	84.78
	No_Time	86.03	84.95	87.11	72.09	93.45	85.88
	No_Transformer	86.27	84.31	88.23	72.60	93.59	86.00
	Ultimate_Fusion_NetWork	86.52	84.58	88.46	73.10	93.86	86.25
6mA_C.elegans	No_BiLSTM	88.38	92.61	84.15	77.04	95.41	88.85
	No_Freq	88.67	92.39	84.95	77.55	95.37	89.07
	No_Time	87.96	91.23	84.69	76.09	94.89	88.35
	No_Transformer	88.38	86.26	90.50	76.83	95.32	88.13
	Ultimate_Fusion_NetWork	91.63	93.47	89.79	83.32	97.08	91.78
6mA_C.equisetifolia	No_BiLSTM	72.29	66.90	77.68	44.84	78.97	70.71
	No_Freq	70.39	75.34	65.44	40.99	78.00	71.79
	No_Time	71.89	74.68	69.10	43.85	80.18	72.65
	No_Transformer	72.22	72.60	71.84	44.45	80.16	72.33
	Ultimate_Fusion_NetWork	74.88	72.63	77.13	49.80	82.45	74.30
6mA_D.melanogaster	No_BiLSTM	89.97	88.44	91.50	79.98	95.65	89.82
	No_Freq	91.23	89.88	92.58	82.50	96.33	91.11
	No_Time	76.32	76.09	76.55	52.64	83.97	76.26
	No_Transformer	90.71	89.85	91.57	81.42	96.42	90.63
	Ultimate_Fusion_NetWork	92.68	91.87	93.49	85.37	97.32	92.62
6mA_F.vesca	No_BiLSTM	90.81	95.29	86.33	81.95	97.04	91.21
	No_Freq	91.78	93.94	89.62	83.64	97.35	91.95
	No_Time	92.10	93.87	90.33	84.26	97.42	92.24
	No_Transformer	91.94	92.84	91.04	83.90	97.32	92.01
	Ultimate_Fusion_NetWork	92.75	93.29	92.21	85.50	97.74	92.79
6mA_H.sapiens	No_BiLSTM	89.17	88.11	90.23	78.36	95.91	89.06
	No_Freq	89.14	86.58	91.70	78.38	96.08	88.86
	No_Time	89.47	88.19	90.75	78.96	96.22	89.33
	No_Transformer	89.64	88.24	91.04	79.32	96.28	89.49
	Ultimate_Fusion_NetWork	90.76	89.04	92.48	81.57	96.93	90.60
6mA_R.chinensis	No_BiLSTM	66.22	73.91	58.53	32.83	72.88	68.63
	No_Freq	72.91	71.24	74.58	45.85	78.89	72.45
	No_Time	69.73	85.62	53.84	41.62	77.56	73.88
	No_Transformer	68.73	78.93	58.53	38.26	77.16	71.62
	Ultimate_Fusion_NetWork	81.77	78.60	84.94	63.67	90.53	81.17
6mA_S.cerevisiae	No_BiLSTM	77.42	70.89	83.95	55.31	86.19	75.84
	No_Freq	78.92	77.81	80.03	57.86	87.59	78.69
	No_Time	77.71	75.22	80.20	55.48	86.34	77.14
	No_Transformer	77.68	70.73	84.63	55.90	86.59	76.01
	Ultimate_Fusion_NetWork	82.44	76.12	88.76	65.39	90.47	81.25
6mA_T.thermophile	No_BiLSTM	87.42	95.27	79.57	75.78	93.85	88.34
	No_Freq	88.23	95.37	81.09	77.26	94.49	89.02
	No_Time	88.29	95.64	80.94	77.43	94.60	89.09
	No_Transformer	88.25	95.58	80.92	77.33	94.51	89.05
	Ultimate_Fusion_NetWork	88.49	95.01	81.97	77.65	94.75	89.20
6mA_Tolypocladium	No_BiLSTM	73.48	76.14	70.82	47.02	81.53	74.16
	No_Freq	73.65	74.84	72.46	47.32	81.05	73.96
	No_Time	73.83	76.08	71.58	47.71	81.20	74.41
	No_Transformer	74.13	78.86	69.40	48.47	82.56	75.30
	Ultimate_Fusion_NetWork	77.47	80.82	74.12	55.07	86.05	78.20
6mA_Xoc BLS256	No_BiLSTM	84.78	89.72	79.84	69.90	92.66	85.50
	No_Freq	88.26	88.35	88.17	76.52	95.07	88.27
	No_Time	87.28	87.12	87.44	74.56	94.41	87.26
	No_Transformer	86.39	85.14	87.64	72.80	93.92	86.22
	Ultimate_Fusion_NetWork	88.90	88.24	89.56	77.80	95.70	88.83